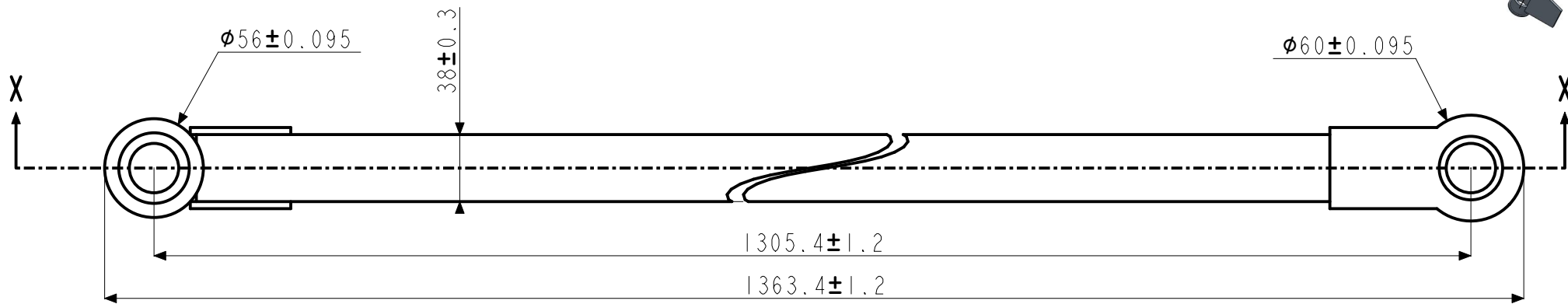
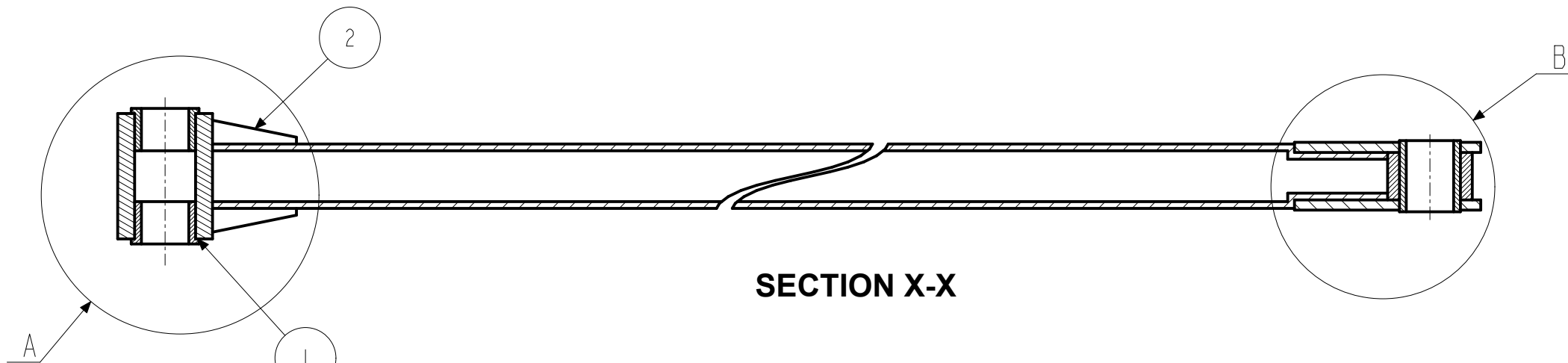


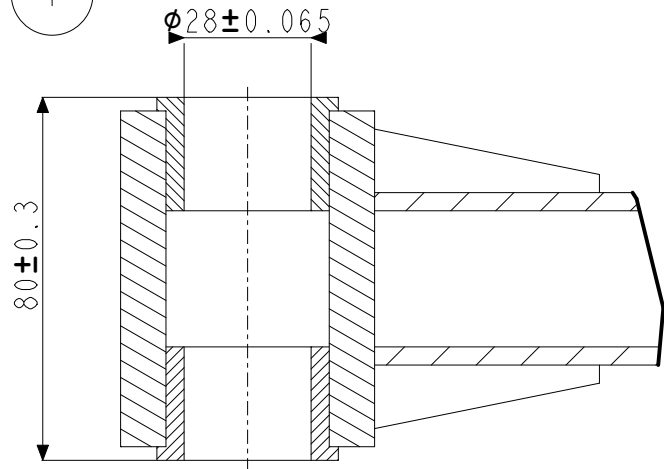
MANIPULATOR ARM



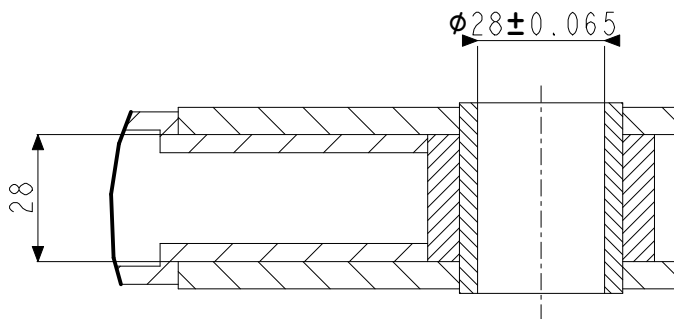
TOP VIEW



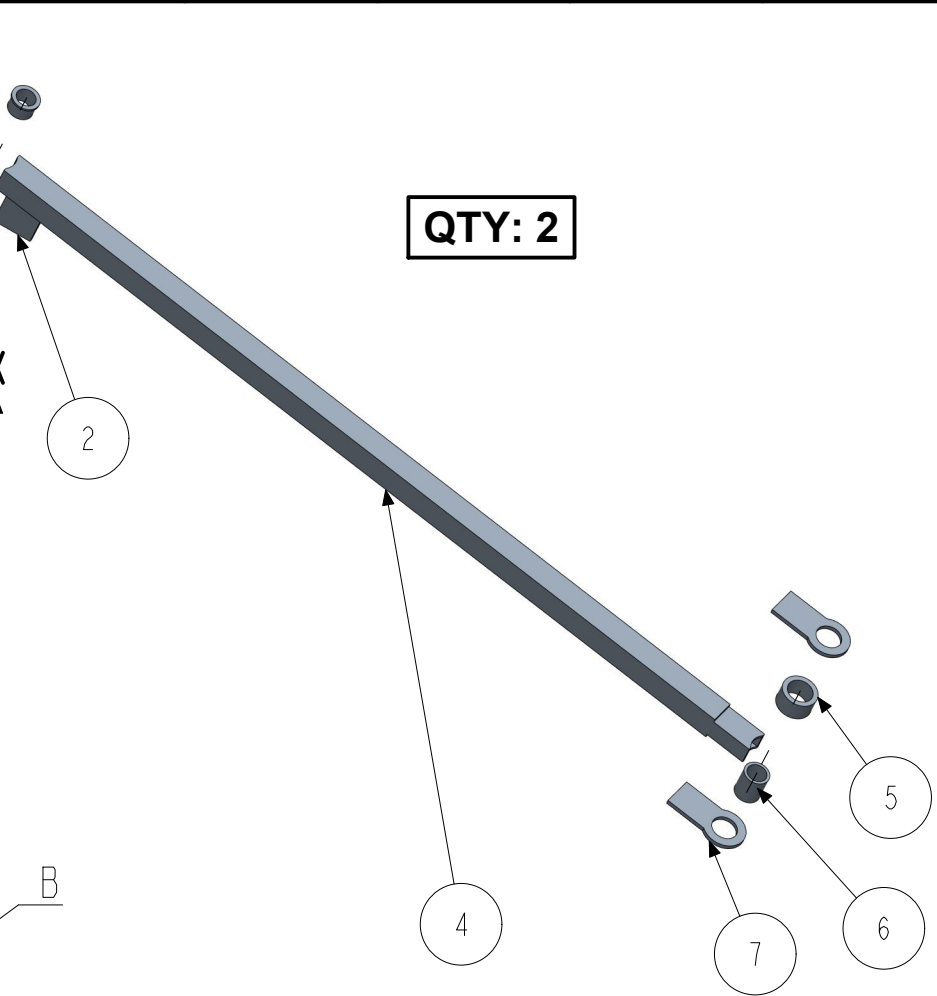
SECTION X-X



DETAIL A
SCALE 0.600



DETAIL B
SCALE 0.600



EXPLODED VIEW

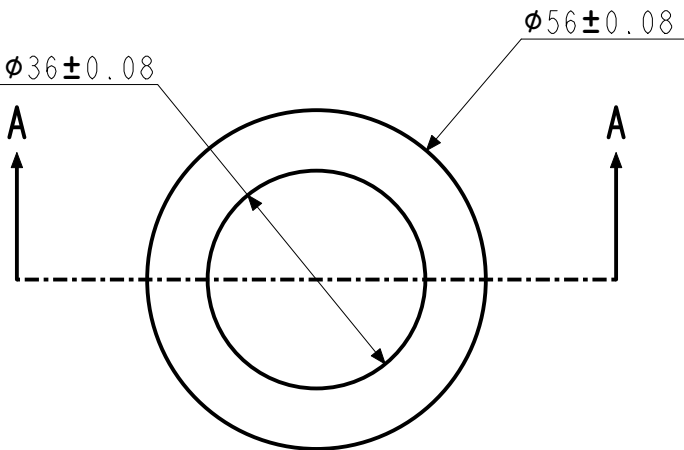
SL. NO.	NAME	QTY
1	02_BUSHING_155-92_SAE64_1	2
2	03_PLATE_156-92_SAE1020_1	2
3	01_BUSHING_154-92_SAE1020_1	1
4	05_PROFILE_158-92_SQURE_TUBE_1	1
5	07_BUSHING_160-92_SAE1020_1	1
6	06_BUSHING_159-92_SAE64_1	1
7	04_PLATE_157-92_SAE1020_1	2

NOTE:

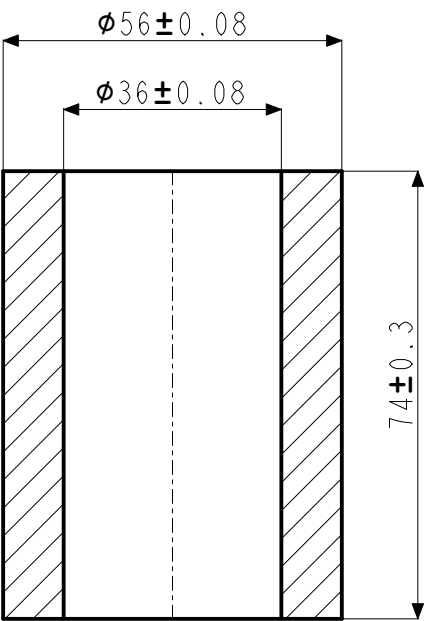
1. ALL DIMENSIONS ARE IN mm
2. Break all sharp edges and remove burrs.
3. General tolerances: $\pm H11/2$ unless otherwise specified.

		DIMENSION 000 - 030 030 - 120 120 - 400 400 - 1000	TOLERANCES ± 0.1 ± 0.2 ± 0.3 ± 0.4			DFAB STAINLESS SYSTEM PVT LTD No: 3B/415, No-8 KIADB Main Road, 1st Stage, 2nd Phase, Peenya Industrial Area, Bengaluru, 560058	
REV	DATE	REMARK		SCALE	1:1	PROJECT	MANIPULATOR ARM
-	-	--		DATE	05-08-2026	DRAWING NO.	DFSSPL-DWRL-0050
				SHEET	1 OF 6	TITLE	GA LAYOUT
				DRWN	AKASH		
				CHKD	LINGRAJ		
MATERIAL				APRD	PRAMOD	CLIENT	ZLAS

BUSHING/ LONG ARM 154-92

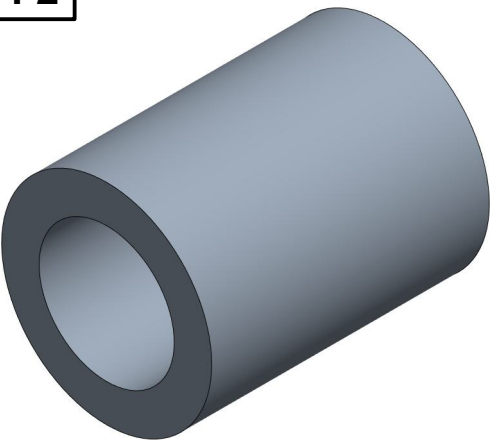


TOP VIEW



SECTION A-A

QTY: 2

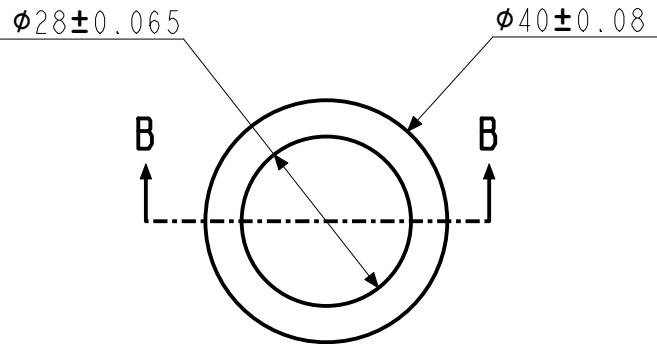


ISOMETRIC VIEW

NOTE:

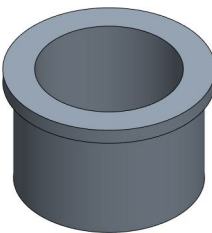
- 1. ALL DIMENSIONS ARE IN mm
- 2. Material: Brass (IS 319 Grade 1 / CuZn37)
- 3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
- 4. Surface finish: N8 unless otherwise specified.
- 5. Break all sharp edges and remove burrs.

BUSHING/ LONG ARM 155-92

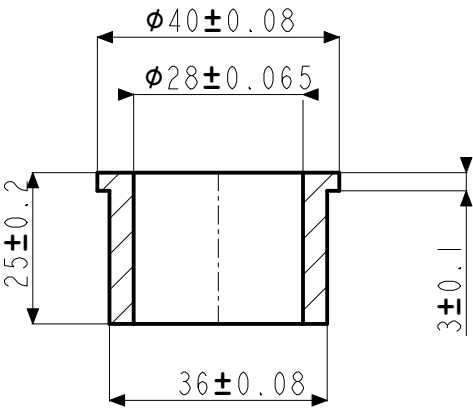


TOP VIEW

QTY: 4



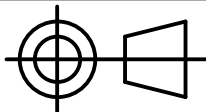
ISOMETRIC VIEW



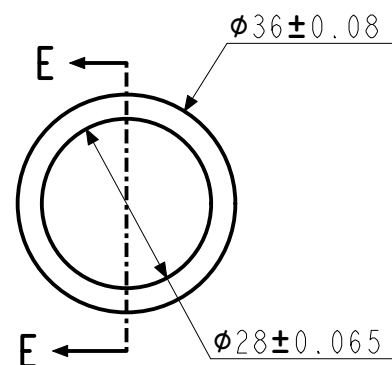
SECTION B-B

NOTE:

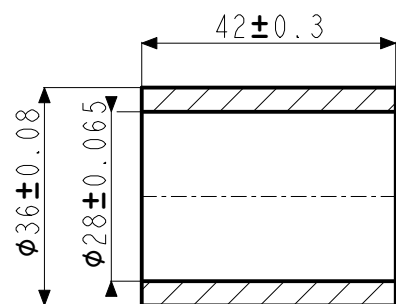
- 1. ALL DIMENSIONS ARE IN mm
- 2. Material: C20 (EN3B/SAE1020 equivalent).
- 3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
- 4. Surface finish: N8 unless otherwise specified.
- 5. Break all sharp edges and remove burrs.

		DIMESION 000 - 030 030 - 120 120 - 400 400 - 1000	TOLERANCES ± 0.1 ± 0.2 ± 0.3 ± 0.4		 DFAB STAINLESS SYSTEM PVT LTD No: 3B/415, No-8 KIADB Main Road, 1st Stage, 2nd Phase,Peenya Industrial Area, Bengaluru, 560058		
REV	DATE	REMARK		SCALE	1:1	PROJECT	MANIPULATOR ARM
-	-	--		DATE	05-08-2026	DRAWING NO.	DFSSPL-DWRL-0050
				SHEET	2 OF 6	TITLE	BUSHING 155-92/ 154-92
				DRWN	AKASH		
				CHKD	LINGRAJ		
MATERIAL				APRD	PRAMOD	CLIENT	ZLAS

BUSHING/ LONG ARM 159-92

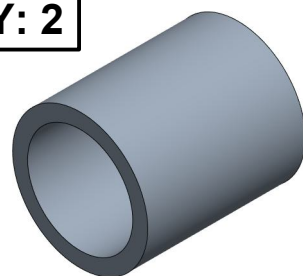


TOP VIEW



SECTION E-E

QTY: 2

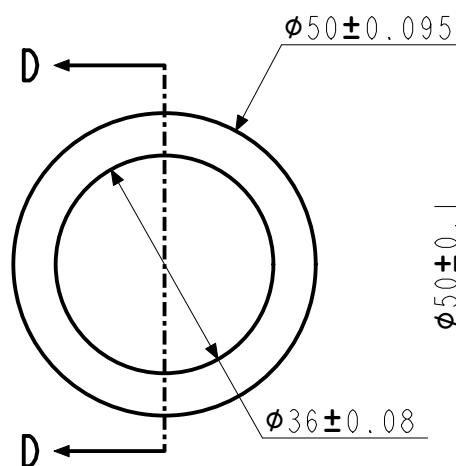


ISOMETRIC VIEW

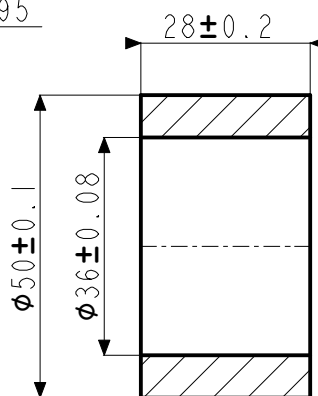
NOTE:

1. ALL DIMENSIONS ARE IN mm
2. Material: Brass (IS 319 Grade 1 / CuZn37)
3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
4. Surface finish: N8 ($R_a \approx 3.2 \mu m$) unless otherwise specified.
5. Finish machine the $\varnothing 28$ bore accurately.
6. Break all sharp edges and remove burrs.

BUSHING/ LONG ARM 160-92

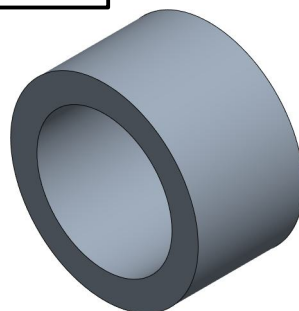


TOP VIEW



SECTION D-D

QTY: 2

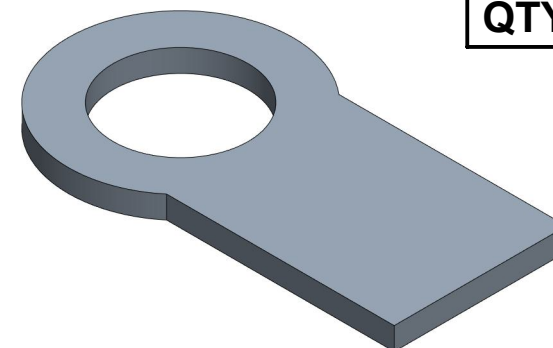


ISOMETRIC VIEW

NOTE:

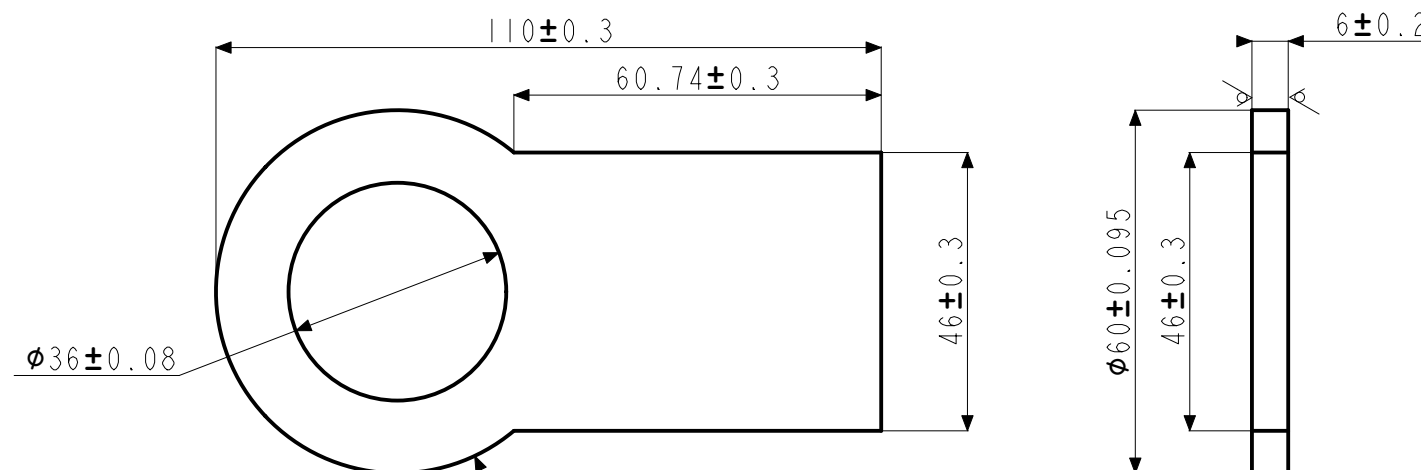
1. ALL DIMENSIONS ARE IN mm
2. Material: C20 (EN3B/SAE1020 equivalent).
3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
4. Surface finish: N8 ($R_a \leq 3.2 \mu m$) unless otherwise specified.
5. Finish machine the $\varnothing 36$ bore accurately.
6. Break all sharp edges and remove burrs.

PLATE 157-92



QTY: 4

ISOMETRIC VIEW



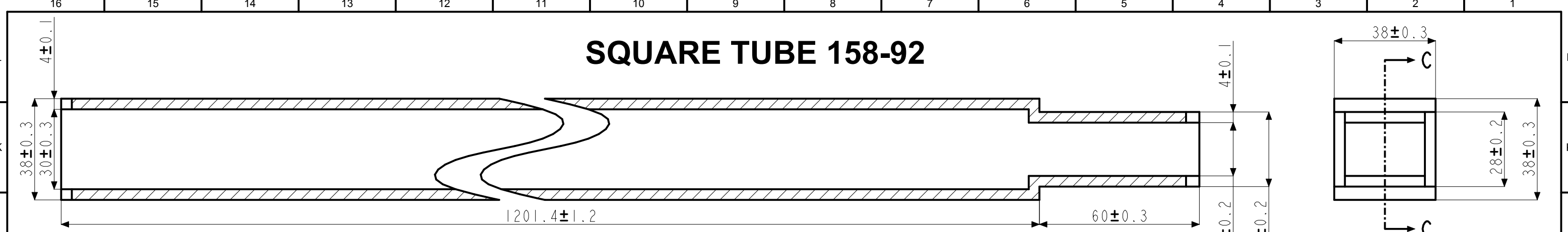
TOP VIEW

SIDE VIEW

NOTE:

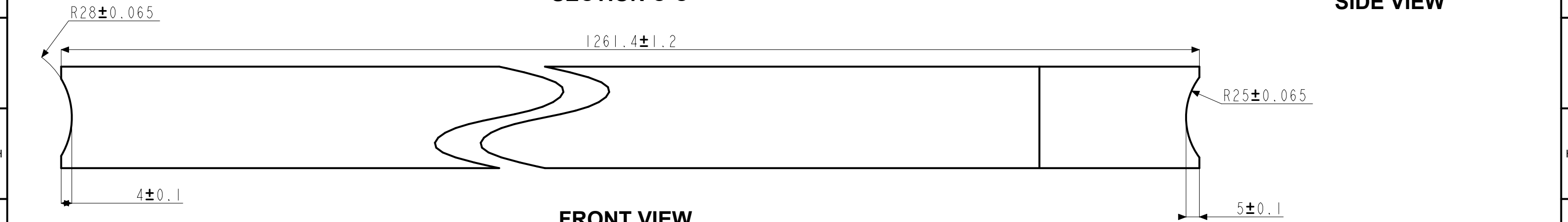
1. ALL DIMENSIONS ARE IN mm
2. Material: IS 2062 E250
3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
4. Surface finish: N8 unless otherwise specified.
5. Break all sharp edges and remove burrs.

		DIMENSION	TOLERANCES			DFAB STAINLESS SYSTEM PVT LTD No: 3B/415, No-8 KIADB Main Road, 1st Stage, 2nd Phase,Peenya Industrial Area, Bengaluru, 560058	
REV	DATE	REMARK		SCALE	1:1	PROJECT	MANIPULATOR ARM
-	-	--		DATE	05-08-2026	DRAWING NO.	DFSSPL-DWRL-0050
				SHEET	3 OF 6	TITLE	BUSHING 159-92/ 160-92/ PLATE 157-92
				DRWN	AKASH		
				CHKD	LINGRAJ		
MATERIAL				APRD	PRAMOD	CLIENT	ZLAS

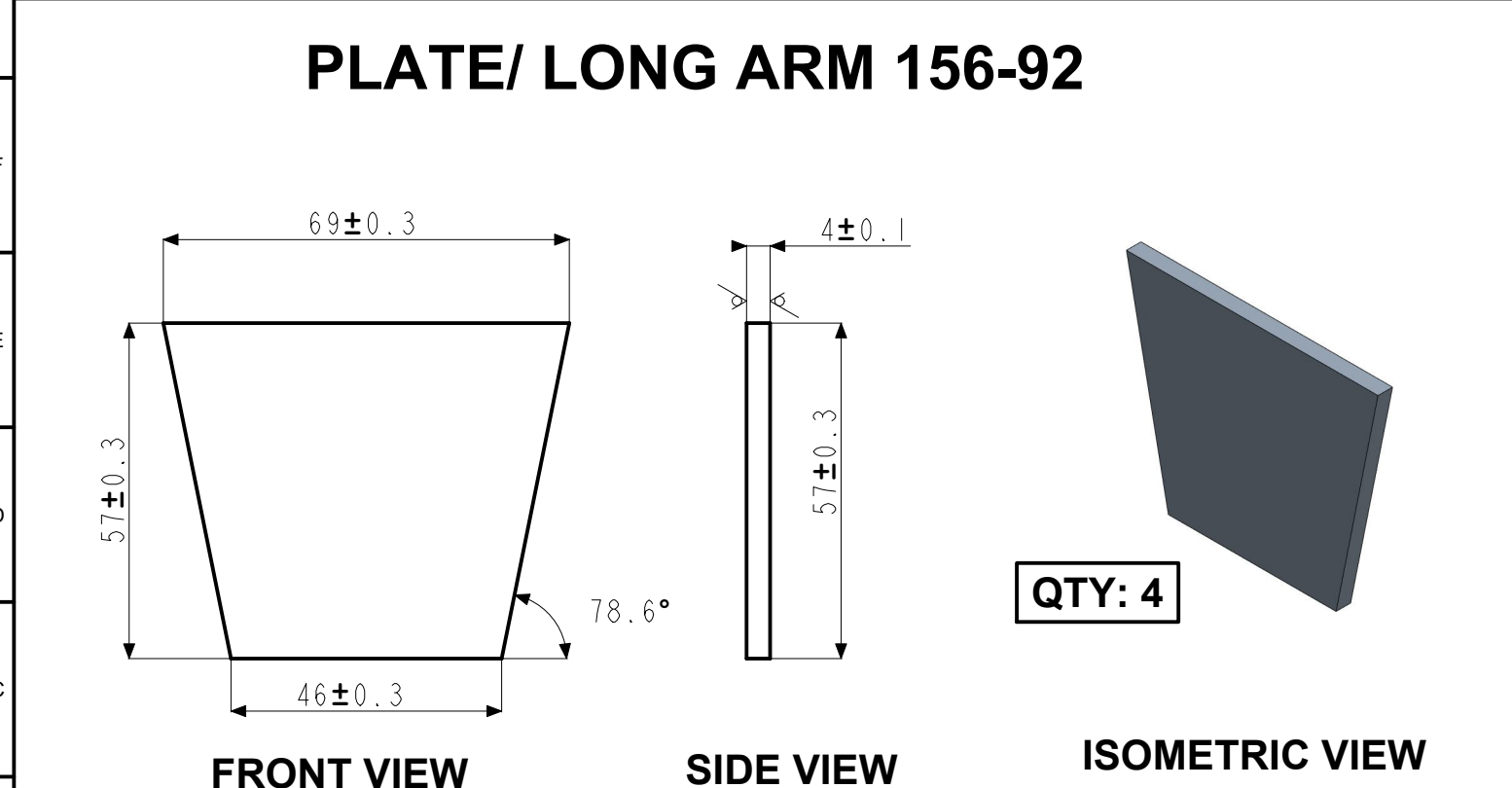


SECTION C-C

SIDE VIEW



FRONT VIEW



QTY: 2

QTY: 4

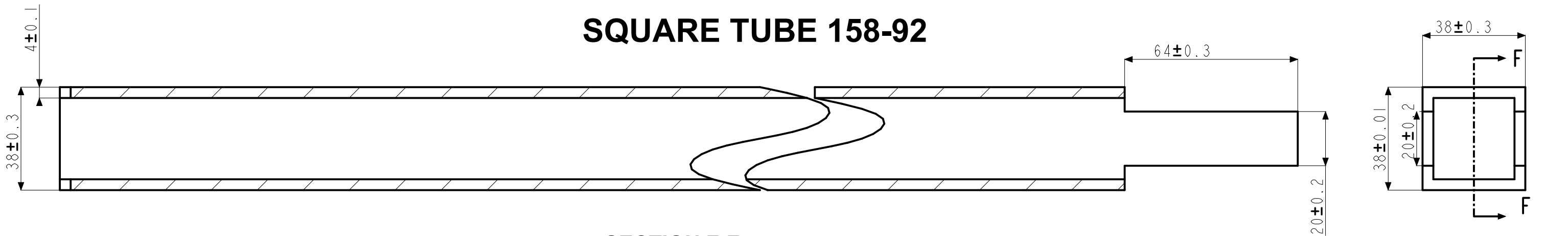
ISOMETRIC VIEW

- NOTE:**
- 1. ALL DIMENSIONS ARE IN mm
 - 2. Material: IS 2062 E250
 - 3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
 - 4. Surface finish: N8 unless otherwise specified.
 - 5. Break all sharp edges and remove burrs.

- NOTE:**
- 1. ALL DIMENSIONS ARE IN mm
 - 2. Material: MS Square Tube (IS 4923 / IS1161)
 - 3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
 - 4. Surface finish: N8 unless otherwise specified.
 - 5. Break all sharp edges and remove burrs.

		DIMESION	TOLERANCES			DFAB STAINLESS SYSTEM PVT LTD No: 3B/415, No-8 KIADB Main Road, 1st Stage, 2nd Phase,Peenya Industrial Area, Bengaluru, 560058	
REV	DATE	REMARK		SCALE	1:1	PROJECT	MANIPULATOR ARM
1.0	07/08/2026	Length increased by 5mm		DATE	05-08-2026	DRAWING NO.	DFSSPL-DWRL-0050
				SHEET	4 OF 6	TITLE	PLATE 156-92/ SQUARE TUBE
				DRWN	AKASH		
				CHKD	LINGRAJ		
MATERIAL				APRD	PRAMOD	CLIENT	ZLAS

SQUARE TUBE 158-92

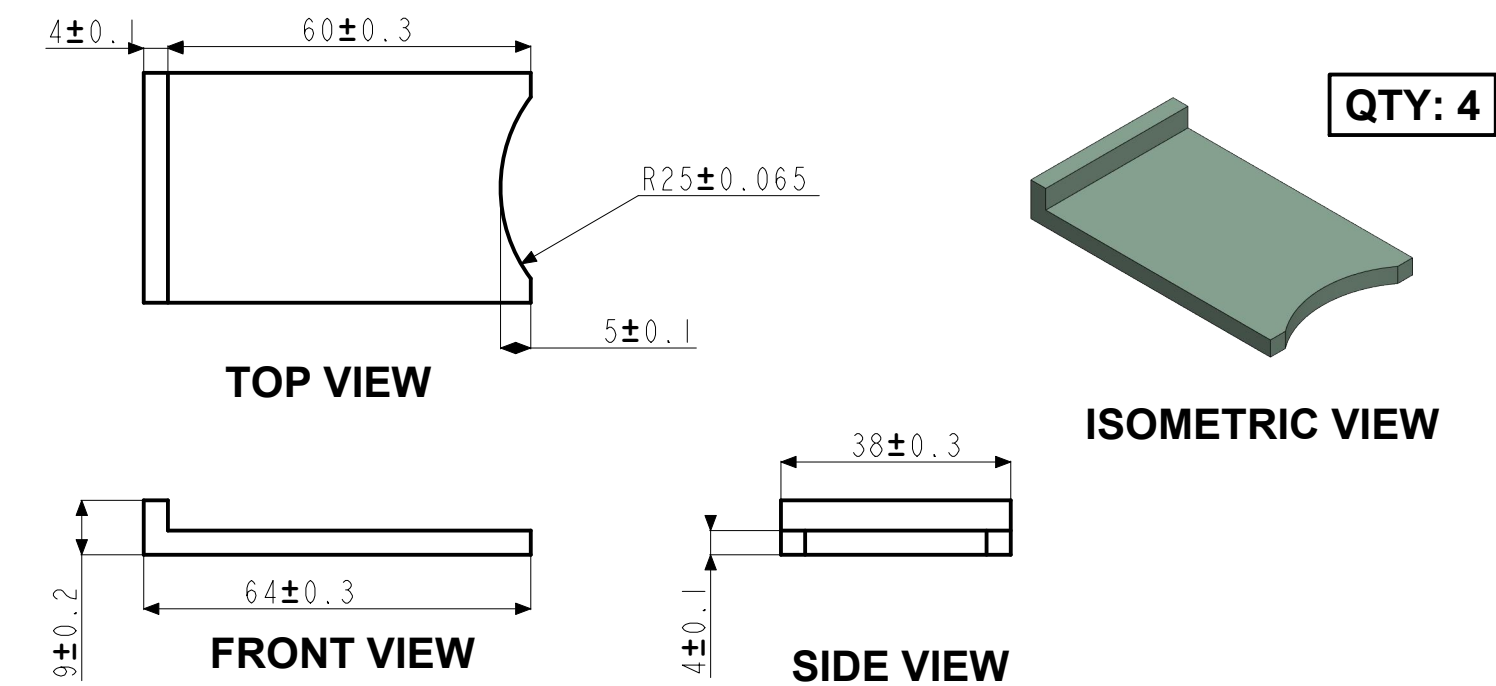


SECTION F-F

SIDE VIEW

FRONT VIEW

SIDE PLATE



TOP VIEW

FRONT VIEW

ISOMETRIC VIEW

SIDE VIEW

QTY: 4

QTY: 2

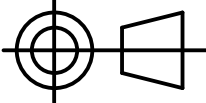
ASSEMBLY

NOTE:

1. ALL DIMENSIONS ARE IN mm
2. Material: MS Square Tube (IS 4923 / IS1161)
3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
4. Surface finish: N8 unless otherwise specified.
5. Break all sharp edges and remove burrs.

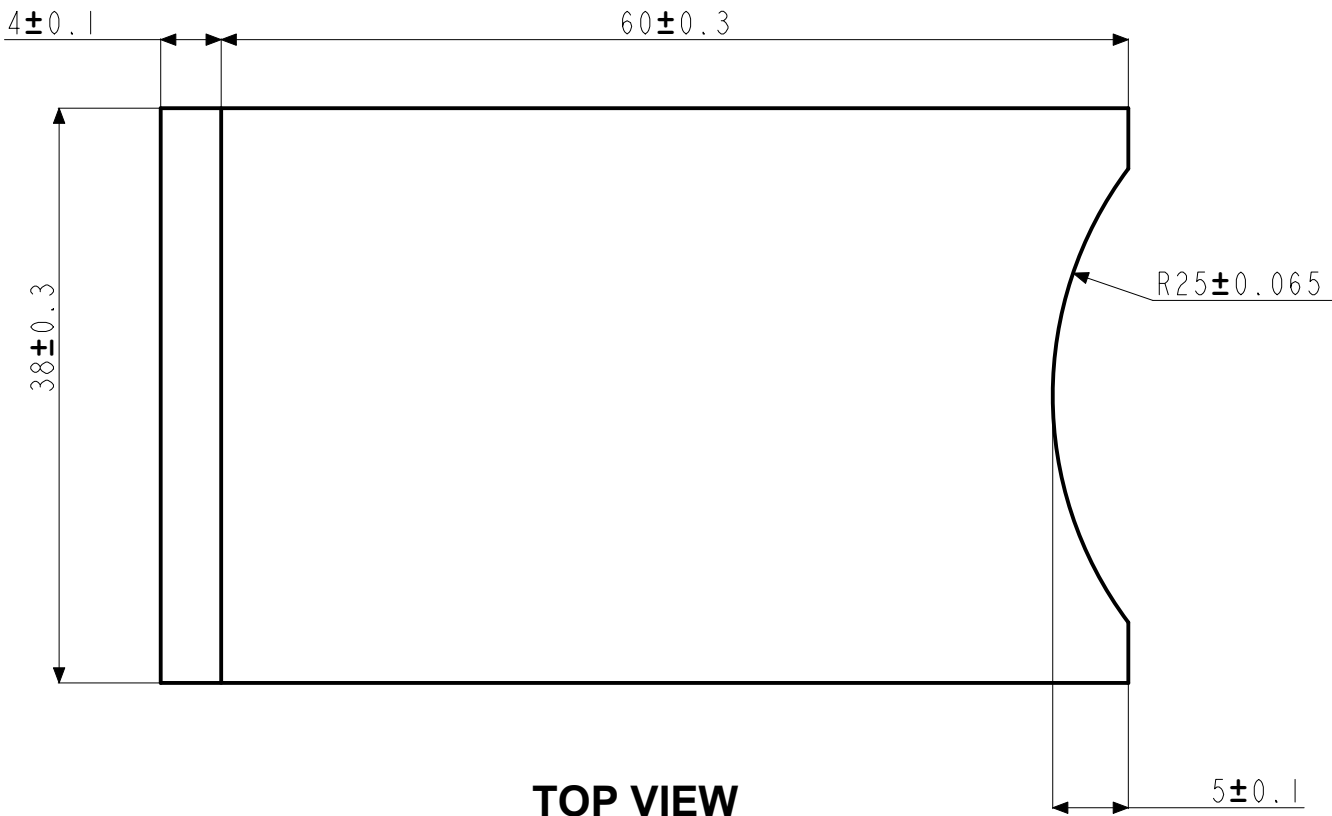
NOTE:

1. ALL DIMENSIONS ARE IN mm
2. Material: IS 4923 / IS1161
3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
4. Surface finish: N8 unless otherwise specified.
5. Break all sharp edges and remove burrs.

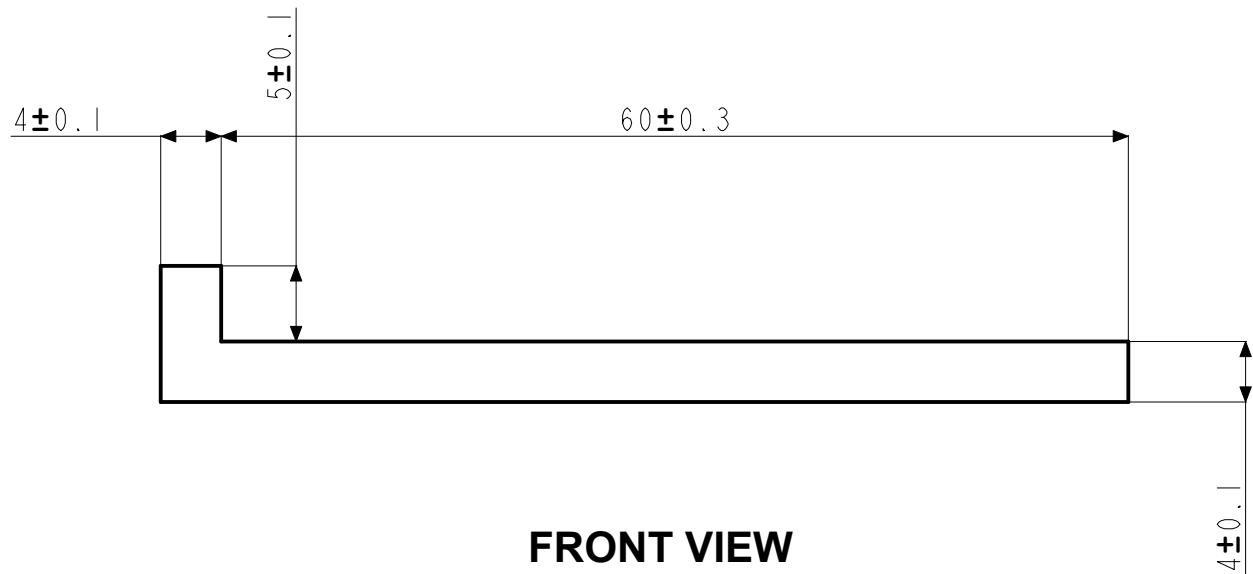
		DIMESION	TOLERANCES			DFAB STAINLESS SYSTEM PVT LTD No: 3B/415, No-8 KIADB Main Road, 1st Stage, 2nd Phase,Peenya Industrial Area, Bengaluru, 560058	
REV	DATE	REMARK		SCALE	1:1	PROJECT	MANIPULATOR ARM
1.0	07/08/2026	Length increased by 5mm		DATE	06-08-2026	DRAWING NO.	DFSSPL-DWRL-0050
				SHEET	5 OF 6	TITLE	SQUARE TUBE ASSEMBLY
				DRWN	AKASH		
				CHKD	LINGRAJ		
MATERIAL				APRD	PRAMOD	CLIENT	ZLAS

SIDE PLATE

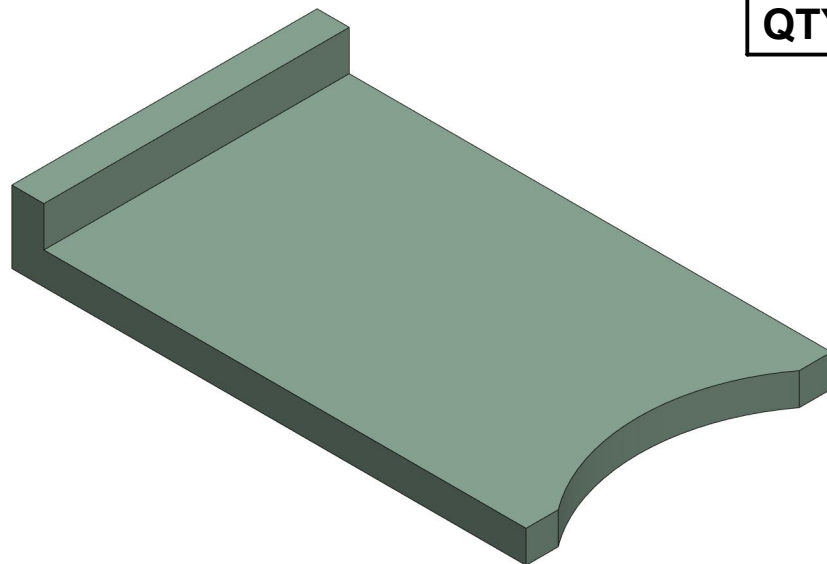
QTY: 4



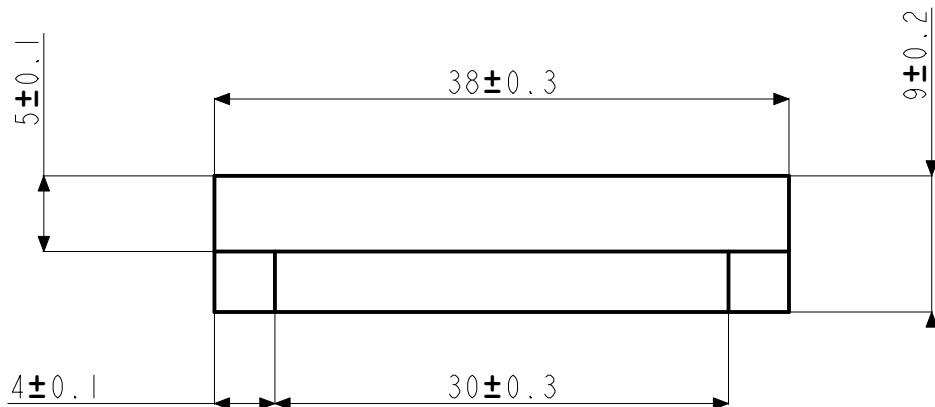
TOP VIEW



FRONT VIEW



ISOMETRIC VIEW



SIDE VIEW

NOTE:

1. ALL DIMENSIONS ARE IN mm
2. Material: IS 4923/ IS 1161
3. Unless otherwise specified, general tolerances shall be as per ISO 2768-m.
4. Surface finish: N8 unless otherwise specified.
5. Break all sharp edges and remove burrs.

		DIMESION 000 - 030 030 - 120 120 - 400 400 - 1000	TOLERANCES ± 0.1 ± 0.2 ± 0.3 ± 0.4			DFAB STAINLESS SYSTEM PVT LTD No: 3B/415, No-8 KIADB Main Road, 1st Stage, 2nd Phase,Peenya Industrial Area, Bengaluru, 560058	
REV	DATE	REMARK		SCALE	1:1	PROJECT	MANIPULATOR ARM
-	-	--		DATE	06-08-2026	DRAWING NO.	DFSSPL-DWRL-0050
				SHEET	6 OF 6	TITLE	SIDE PLATE
				DRWN	AKASH		
				CHKD	LINGRAJ		
MATERIAL				APRD	PRAMOD	CLIENT	ZLAS